COMP4033 G5_A1 - Sleep

Group 5

2026-03-26

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)
library(dplyr)
```

```r
sleepDS <- read.csv("Sleep_health_and_lifestyle_dataset.csv")
#Please upload the dataset!
```

#Understanding the dataset

```r
#1. Print the structure of your dataset.
str(sleepDS)
```

```
## 'data.frame':    374 obs. of  13 variables:
##  $ Person.ID              : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Gender                 : chr  "Male" "Male" "Male" "Male" ...
##  $ Age                    : int  27 28 28 28 28 28 29 29 29 29 ...
##  $ Occupation             : chr  "Software Engineer" "Doctor" "Doctor" "Sales Representative" ...
##  $ Sleep.Duration         : num  6.1 6.2 6.2 5.9 5.9 5.9 6.3 7.8 7.8 7.8 ...
##  $ Quality.of.Sleep       : int  6 6 6 4 4 4 6 7 7 7 ...
##  $ Physical.Activity.Level: int  42 60 60 30 30 30 40 75 75 75 ...
##  $ Stress.Level           : int  6 8 8 8 8 8 7 6 6 6 ...
##  $ BMI.Category           : chr  "Overweight" "Normal" "Normal" "Obese" ...
##  $ Blood.Pressure         : chr  "126/83" "125/80" "125/80" "140/90" ...
##  $ Heart.Rate             : int  77 75 75 85 85 85 82 70 70 70 ...
##  $ Daily.Steps            : int  4200 10000 10000 3000 3000 3000 3500 8000 8000 8000 ...
##  $ Sleep.Disorder         : chr  "None" "None" "None" "Sleep Apnea" ...
```

```r
#2. List the variables in your dataset.
ls(sleepDS)
```

```
##  [1] "Age"                      "Blood.Pressure"
##  [3] "BMI.Category"             "Daily.Steps"
##  [5] "Gender"                   "Heart.Rate"
```

```
##  [7] "Occupation"              "Person.ID"
##  [9] "Physical.Activity.Level" "Quality.of.Sleep"
## [11] "Sleep.Disorder"          "Sleep.Duration"
## [13] "Stress.Level"
```

```r
#3. Print the top 15 rows of your dataset.
head(sleepDS, n=15)
```

```
##    Person.ID Gender Age           Occupation Sleep.Duration Quality.of.Sleep
## 1          1   Male  27    Software Engineer            6.1                6
## 2          2   Male  28               Doctor            6.2                6
## 3          3   Male  28               Doctor            6.2                6
## 4          4   Male  28 Sales Representative            5.9                4
## 5          5   Male  28 Sales Representative            5.9                4
## 6          6   Male  28    Software Engineer            5.9                4
## 7          7   Male  29              Teacher            6.3                6
## 8          8   Male  29               Doctor            7.8                7
## 9          9   Male  29               Doctor            7.8                7
## 10        10   Male  29               Doctor            7.8                7
## 11        11   Male  29               Doctor            6.1                6
## 12        12   Male  29               Doctor            7.8                7
## 13        13   Male  29               Doctor            6.1                6
## 14        14   Male  29               Doctor            6.0                6
## 15        15   Male  29               Doctor            6.0                6
##    Physical.Activity.Level Stress.Level BMI.Category Blood.Pressure Heart.Rate
## 1                       42            6   Overweight         126/83         77
## 2                       60            8       Normal         125/80         75
## 3                       60            8       Normal         125/80         75
## 4                       30            8        Obese         140/90         85
## 5                       30            8        Obese         140/90         85
## 6                       30            8        Obese         140/90         85
## 7                       40            7        Obese         140/90         82
## 8                       75            6       Normal         120/80         70
## 9                       75            6       Normal         120/80         70
## 10                      75            6       Normal         120/80         70
## 11                      30            8       Normal         120/80         70
## 12                      75            6       Normal         120/80         70
## 13                      30            8       Normal         120/80         70
## 14                      30            8       Normal         120/80         70
## 15                      30            8       Normal         120/80         70
##    Daily.Steps Sleep.Disorder
## 1         4200           None
## 2        10000           None
## 3        10000           None
## 4         3000    Sleep Apnea
## 5         3000    Sleep Apnea
## 6         3000       Insomnia
## 7         3500       Insomnia
## 8         8000           None
## 9         8000           None
## 10        8000           None
## 11        8000           None
## 12        8000           None
## 13        8000           None
## 14        8000           None
```

```
## 15        8000            None
```

Upon the exploration of a chosen dataset as the first step, called function shows the dataset renamed as "sleepDS" contains 13 variables and 374 values. The data types of each attribute vary according to the values, including integer, character, and numeric.

#Cleaning dataset

```r
#4. Remove missing values in your dataset.
summary(is.na(sleepDS))
```

```
##  Person.ID         Gender           Age          Occupation
##  Mode :logical   Mode :logical   Mode :logical   Mode :logical
##  FALSE:374       FALSE:374       FALSE:374       FALSE:374
##  Sleep.Duration  Quality.of.Sleep Physical.Activity.Level Stress.Level
##  Mode :logical   Mode :logical    Mode :logical           Mode :logical
##  FALSE:374       FALSE:374        FALSE:374               FALSE:374
##  BMI.Category    Blood.Pressure  Heart.Rate      Daily.Steps
##  Mode :logical   Mode :logical   Mode :logical   Mode :logical
##  FALSE:374       FALSE:374       FALSE:374       FALSE:374
##  Sleep.Disorder
##  Mode :logical
##  FALSE:374
```

Missing value is not identified from the above function, hence, no need to perform the removal/replacement of missing value.This can be seen in the summary of the is.na function where FALSE is returned for all 374 observations across the 13 variables.

```r
#5. Identify and remove duplicated data from your dataset.
summary(duplicated(sleepDS))
```

```
##    Mode   FALSE
## logical     374
```

```r
sleepDS_dup = sleepDS %>% distinct()
str(sleepDS_dup)
```

```
## 'data.frame':    374 obs. of  13 variables:
##  $ Person.ID              : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Gender                 : chr  "Male" "Male" "Male" "Male" ...
##  $ Age                    : int  27 28 28 28 28 28 29 29 29 29 ...
##  $ Occupation             : chr  "Software Engineer" "Doctor" "Doctor" "Sales Representative" ...
##  $ Sleep.Duration         : num  6.1 6.2 6.2 5.9 5.9 5.9 6.3 7.8 7.8 7.8 ...
##  $ Quality.of.Sleep       : int  6 6 6 4 4 4 6 7 7 7 ...
##  $ Physical.Activity.Level: int  42 60 60 30 30 30 40 75 75 75 ...
##  $ Stress.Level           : int  6 8 8 8 8 8 7 6 6 6 ...
##  $ BMI.Category           : chr  "Overweight" "Normal" "Normal" "Obese" ...
##  $ Blood.Pressure         : chr  "126/83" "125/80" "125/80" "140/90" ...
##  $ Heart.Rate             : int  77 75 75 85 85 85 82 70 70 70 ...
##  $ Daily.Steps            : int  4200 10000 10000 3000 3000 3000 3500 8000 8000 8000 ...
##  $ Sleep.Disorder         : chr  "None" "None" "None" "Sleep Apnea" ...
```

No duplicate value identified new data.frame sleepDS.dup created with 374 obj and 13 variables just as parent sleep.DS.

```r
#6. Rename some of the column names in your dataset.
sleepDS_cl <- sleepDS_dup %>% rename(
  Sleep = Sleep.Duration,
  Stress = Stress.Level,
```

```r
  Physical.Activity = Physical.Activity.Level)
str(sleepDS_cl)
```

```
## 'data.frame':    374 obs. of  13 variables:
##  $ Person.ID        : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Gender           : chr  "Male" "Male" "Male" "Male" ...
##  $ Age              : int  27 28 28 28 28 28 29 29 29 29 ...
##  $ Occupation       : chr  "Software Engineer" "Doctor" "Doctor" "Sales Representative" ...
##  $ Sleep            : num  6.1 6.2 6.2 5.9 5.9 5.9 6.3 7.8 7.8 7.8 ...
##  $ Quality.of.Sleep : int  6 6 6 4 4 4 6 7 7 7 ...
##  $ Physical.Activity: int  42 60 60 30 30 30 40 75 75 75 ...
##  $ Stress           : int  6 8 8 8 8 8 7 6 6 6 ...
##  $ BMI.Category     : chr  "Overweight" "Normal" "Normal" "Obese" ...
##  $ Blood.Pressure   : chr  "126/83" "125/80" "125/80" "140/90" ...
##  $ Heart.Rate       : int  77 75 75 85 85 85 82 70 70 70 ...
##  $ Daily.Steps      : int  4200 10000 10000 3000 3000 3000 3500 8000 8000 8000 ...
##  $ Sleep.Disorder   : chr  "None" "None" "None" "Sleep Apnea" ...
```

The data.frame is converted into a new frame sleepDS.cl with renamed variable of sleep,stress and Physical.Activity for legibility. #Data Manipulation

```r
# 7.    Use data manipulation techniques and filter rows based on any logical
# criteria that exist in your dataset.
sleepDS_fil = sleepDS_cl %>% filter(Age >=30)
head(sleepDS_fil %>% arrange(Age))
```

```
##   Person.ID Gender Age Occupation Sleep Quality.of.Sleep Physical.Activity
## 1        20   Male  30     Doctor   7.6                7                75
## 2        21   Male  30     Doctor   7.7                7                75
## 3        22   Male  30     Doctor   7.7                7                75
## 4        23   Male  30     Doctor   7.7                7                75
## 5        24   Male  30     Doctor   7.7                7                75
## 6        25   Male  30     Doctor   7.8                7                75
##   Stress BMI.Category Blood.Pressure Heart.Rate Daily.Steps Sleep.Disorder
## 1      6       Normal         120/80         70        8000           None
## 2      6       Normal         120/80         70        8000           None
## 3      6       Normal         120/80         70        8000           None
## 4      6       Normal         120/80         70        8000           None
## 5      6       Normal         120/80         70        8000           None
## 6      6       Normal         120/80         70        8000           None
```

Due to the incomplete Gen Z value, the subjects were filtered by the age to focus on those who are Gen Y (Millennial) and Gen X. These are people 30 years or older in 2026.

```r
#8. Reorder multiple rows in descending order
sleepDS_ord = sleepDS_fil %>% arrange(desc(Age))
head(sleepDS_ord)
```

```
##   Person.ID Gender Age Occupation Sleep Quality.of.Sleep Physical.Activity
## 1       359 Female  59      Nurse   8.0                9                75
## 2       360 Female  59      Nurse   8.1                9                75
## 3       361 Female  59      Nurse   8.2                9                75
## 4       362 Female  59      Nurse   8.2                9                75
## 5       363 Female  59      Nurse   8.2                9                75
## 6       364 Female  59      Nurse   8.2                9                75
##   Stress BMI.Category Blood.Pressure Heart.Rate Daily.Steps Sleep.Disorder
```

```
## 1       3   Overweight         140/95         68        7000           None
## 2       3   Overweight         140/95         68        7000           None
## 3       3   Overweight         140/95         68        7000    Sleep Apnea
## 4       3   Overweight         140/95         68        7000    Sleep Apnea
## 5       3   Overweight         140/95         68        7000    Sleep Apnea
## 6       3   Overweight         140/95         68        7000    Sleep Apnea
```

Objects have now been organized by a descending order in a new data frame sleepDS_ord

```
#9. Identify the dependent & independent variables
dependent = c("Quality of Sleep")
independent = c("Age", "Sleep", "Physical.Activity", "Stress", "Heart.Rate",
                "Daily.Steps", "GenderBI")
dependent
```

```
## [1] "Quality of Sleep"
```

```
independent #all of the quantity attributes
```

```
## [1] "Age"               "Sleep"             "Physical.Activity"
## [4] "Stress"            "Heart.Rate"        "Daily.Steps"
## [7] "GenderBI"
```

New values listing out dependent and independent variables were created.

```
#Assign the binary values to Gender for future analysis.
sleepDS_ord$GenderBI = ifelse(sleepDS_ord$Gender == "Male", 1, 0)
str(sleepDS_ord)
```

```
## 'data.frame':    355 obs. of  14 variables:
##  $ Person.ID        : int  359 360 361 362 363 364 365 366 367 368 ...
##  $ Gender           : chr  "Female" "Female" "Female" "Female" ...
##  $ Age              : int  59 59 59 59 59 59 59 59 59 59 ...
##  $ Occupation       : chr  "Nurse" "Nurse" "Nurse" "Nurse" ...
##  $ Sleep            : num  8 8.1 8.2 8.2 8.2 8.2 8 8 8.1 8 ...
##  $ Quality.of.Sleep : int  9 9 9 9 9 9 9 9 9 9 ...
##  $ Physical.Activity: int  75 75 75 75 75 75 75 75 75 75 ...
##  $ Stress           : int  3 3 3 3 3 3 3 3 3 3 ...
##  $ BMI.Category     : chr  "Overweight" "Overweight" "Overweight" "Overweight" ...
##  $ Blood.Pressure   : chr  "140/95" "140/95" "140/95" "140/95" ...
##  $ Heart.Rate       : int  68 68 68 68 68 68 68 68 68 68 ...
##  $ Daily.Steps      : int  7000 7000 7000 7000 7000 7000 7000 7000 7000 7000 ...
##  $ Sleep.Disorder   : chr  "None" "None" "Sleep Apnea" "Sleep Apnea" ...
##  $ GenderBI         : num  0 0 0 0 0 0 0 0 0 0 ...
```

A new variable is added to sleepDS_ord assigning gender a numeric value for analysis.

```
#9. and use reshaping techniques and create a new data frame by joining
#those variables from your dataset.
sleepDF = sleepDS_ord %>% select(Quality.of.Sleep,
                                 GenderBI, Age, Sleep, Physical.Activity,
                                 Stress, Heart.Rate, Daily.Steps)
class(sleepDF)
```

```
## [1] "data.frame"
```

```
str(sleepDF)
```

```
## 'data.frame':    355 obs. of  8 variables:
```

```
##  $ Quality.of.Sleep : int  9 9 9 9 9 9 9 9 9 9 ...
##  $ GenderBI         : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ Age              : int  59 59 59 59 59 59 59 59 59 59 ...
##  $ Sleep            : num  8 8.1 8.2 8.2 8.2 8.2 8 8 8.1 8 ...
##  $ Physical.Activity: int  75 75 75 75 75 75 75 75 75 75 ...
##  $ Stress           : int  3 3 3 3 3 3 3 3 3 3 ...
##  $ Heart.Rate       : int  68 68 68 68 68 68 68 68 68 68 ...
##  $ Daily.Steps      : int  7000 7000 7000 7000 7000 7000 7000 7000 7000 7000 ...
```

The new data frame focusing only on our independent and dependent variables was created. This data frame named "sleepDF" is cleaned with no missing and duplicated values as per function 4. and 5. In this new data frame, there are 8 variables and 355 values.

```r
#10. Add new variables in your data frame by using a mathematical function
sleepDF$Steps = (sleepDF$Daily.Steps)/1000
str(sleepDF)
```

```
## 'data.frame':    355 obs. of  9 variables:
##  $ Quality.of.Sleep : int  9 9 9 9 9 9 9 9 9 9 ...
##  $ GenderBI         : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ Age              : int  59 59 59 59 59 59 59 59 59 59 ...
##  $ Sleep            : num  8 8.1 8.2 8.2 8.2 8.2 8 8 8.1 8 ...
##  $ Physical.Activity: int  75 75 75 75 75 75 75 75 75 75 ...
##  $ Stress           : int  3 3 3 3 3 3 3 3 3 3 ...
##  $ Heart.Rate       : int  68 68 68 68 68 68 68 68 68 68 ...
##  $ Daily.Steps      : int  7000 7000 7000 7000 7000 7000 7000 7000 7000 7000 ...
##  $ Steps            : num  7 7 7 7 7 7 7 7 7 7 ...
```

To reduce the data dimension, the daily.steps were changed from "steps unit" to "k steps unit".

#Analysis

```r
#11. Print the summary statistics of your dataset.
summary(sleepDF)
```

```
##  Quality.of.Sleep    GenderBI           Age            Sleep
##  Min.   :4.000    Min.   :0.0000   Min.   :30.00   Min.   :5.800
##  1st Qu.:6.000    1st Qu.:0.0000   1st Qu.:37.00   1st Qu.:6.500
##  Median :8.000    Median :0.0000   Median :43.00   Median :7.200
##  Mean   :7.394    Mean   :0.4845   Mean   :42.91   Mean   :7.168
##  3rd Qu.:8.000    3rd Qu.:1.0000   3rd Qu.:50.00   3rd Qu.:7.800
##  Max.   :9.000    Max.   :1.0000   Max.   :59.00   Max.   :8.500
##  Physical.Activity     Stress        Heart.Rate     Daily.Steps
##  Min.   :30.00     Min.   :3.000   Min.   :65.00   Min.   : 3300
##  1st Qu.:45.00     1st Qu.:4.000   1st Qu.:68.00   1st Qu.: 6000
##  Median :60.00     Median :5.000   Median :70.00   Median : 7000
##  Mean   :59.94     Mean   :5.282   Mean   :69.91   Mean   : 6830
##  3rd Qu.:75.00     3rd Qu.:7.000   3rd Qu.:72.00   3rd Qu.: 8000
##  Max.   :90.00     Max.   :8.000   Max.   :86.00   Max.   :10000
##      Steps
##  Min.   : 3.30
##  1st Qu.: 6.00
##  Median : 7.00
##  Mean   : 6.83
##  3rd Qu.: 8.00
##  Max.   :10.00
```

The summary function applies some basic mathematical functions to each variable, but these and more

functions are done separatly for a more complete analysis.

```
#12.    Write a user defined function using any of the variables from the data set.
Find_Mode = function(x){
  ux=unique(x)
  ux[which.max(tabulate(match(x,ux)))]}
test = c(1,1,2,5,2,5,1,1,0)
Find_Mode(test)
```

```
## [1] 1
```

Creation of a function to help identify the mode of our variables.

```
#13.    Use any of the numerical variables from the dataset and perform
#the following statistical functions.
#Mean
mean(sleepDF$Quality.of.Sleep)
```

```
## [1] 7.394366
```

```
#13.    Use any of the numerical variables from the dataset and perform
#the following statistical functions.
#Median
median(sleepDF$Quality.of.Sleep)
```

```
## [1] 8
```

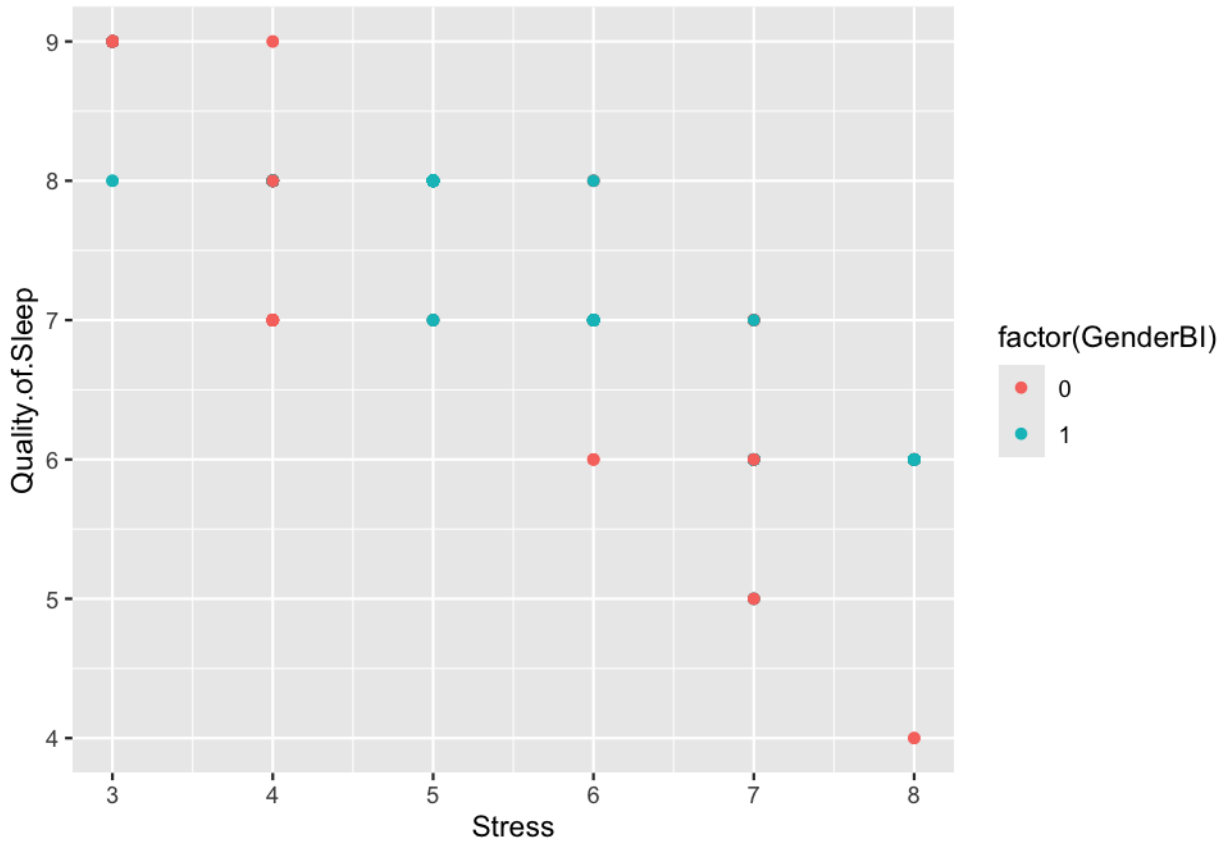
```
#13.    Use any of the numerical variables from the dataset and perform
#the following statistical functions.
#Mode
Find_Mode(sleepDF$Quality.of.Sleep)
```

```
## [1] 8
```

```
#13.    Use any of the numerical variables from the dataset and performthe
# following statistical functions.
#Range
range(sleepDF$Quality.of.Sleep, na.rm=TRUE)
```

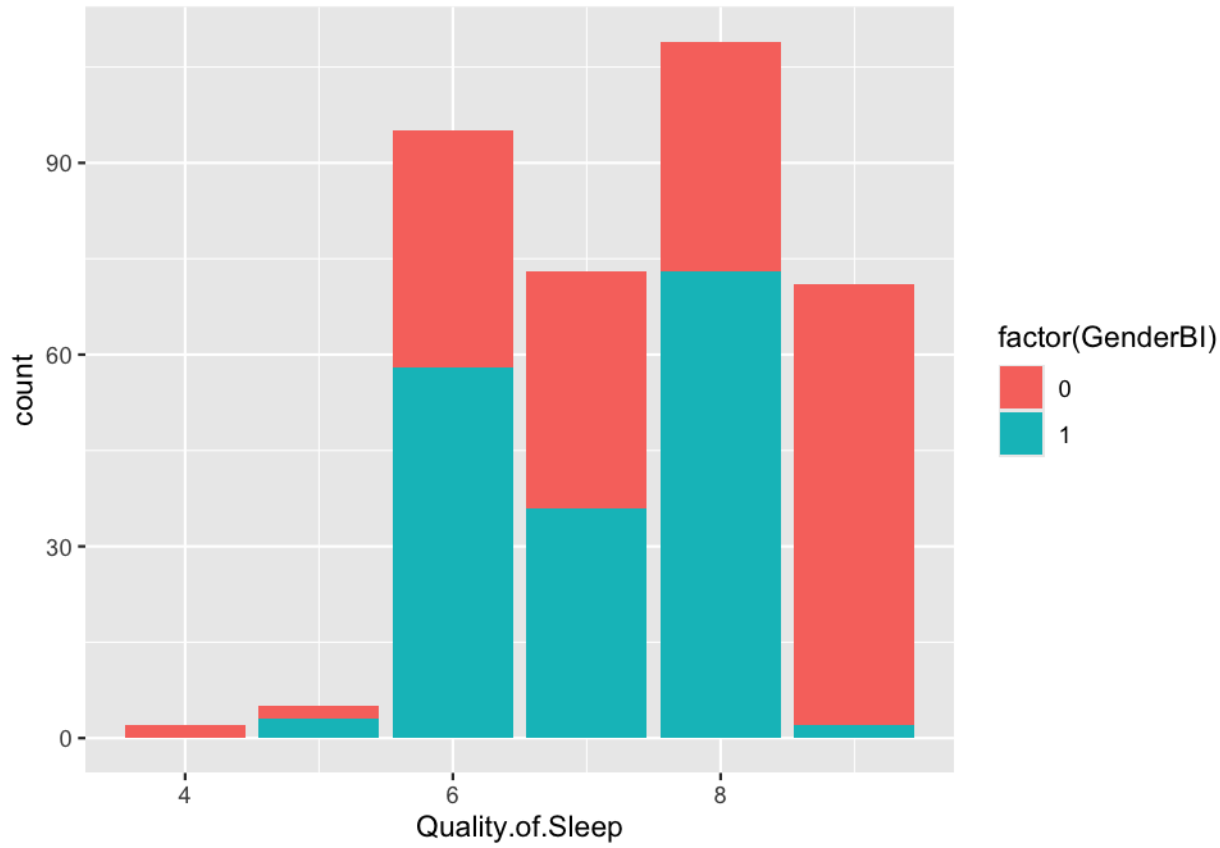
```
## [1] 4 9
```

```
#14.    Plot a scatter plot for any 2 variables in your dataset.
ggplot(data= sleepDF,
       aes(x=Stress, y=Quality.of.Sleep, col=factor(GenderBI)))+geom_point()
```



The scatter plot suggests an overall negative relationship between stress level and quality of sleep. When stress is low (3-5), the quality of sleep tends to be around the high score area (7-9); whereas with higher stress level (7-8), lower sleep qiality scores are observed (4-6). The pattern indicates that increasing stress level may be associated with poorer quality of sleep. As for the two gender groups, there is no clear visual evidence of gender specific pattern with stress level or quality of sleep based on the plot.

```
#15.    Plot a bar plot for any 2 variables in your dataset
ggplot(data=sleepDF,
       aes(x=Quality.of.Sleep, fill=factor(GenderBI)))+geom_bar()
```



The bar chart shows that the data are concentrated in the mid to high sleep quality range (6-9), suggesting that most participants selected reported moderate to good sleep quality. Across scores 6-8, both gender groups are represented and appear fairly balanced, with the Male group (GenderBI=1) slightly higher. At the highest of sleep quality score, Female group (GenderBI=0) accounts for the most of responses indicating that gender may be associated with difference in distribution of sleep quality rating. However, the chart alone does not confirm the statistical significance, so further testing or data analysis are needed to validate the pattern.

```
#16.    Find the correlation between any 2 variables by applying least square
# linear regression model.
cor_matrix = cor(sleepDF)
cor_matrix
```

```
##                    Quality.of.Sleep    GenderBI         Age       Sleep
## Quality.of.Sleep          1.0000000 -0.26837214  0.41818352  0.88951697
## GenderBI                 -0.2683721  1.00000000 -0.58464846 -0.09231296
## Age                       0.4181835 -0.58464846  1.00000000  0.30565771
## Sleep                     0.8895170 -0.09231296  0.30565771  1.00000000
## Physical.Activity         0.1346722  0.02936620  0.13416027  0.15739845
## Stress                   -0.9013346  0.37240079 -0.36931413 -0.80538274
## Heart.Rate               -0.6237316  0.20556299 -0.14884805 -0.50233944
## Daily.Steps              -0.0436656 -0.02720559  0.04924485 -0.07307288
## Steps                    -0.0436656 -0.02720559  0.04924485 -0.07307288
##                    Physical.Activity      Stress  Heart.Rate Daily.Steps
## Quality.of.Sleep          0.13467221 -0.90133465 -0.62373160 -0.04366560
## GenderBI                  0.02936620  0.37240079  0.20556299 -0.02720559
## Age                       0.13416027 -0.36931413 -0.14884805  0.04924485
## Sleep                     0.15739845 -0.80538274 -0.50233944 -0.07307288
```

```
## Physical.Activity        1.00000000  0.02371081  0.22872099  0.81117060
## Stress                   0.02371081  1.00000000  0.68187966  0.21514012
## Heart.Rate               0.22872099  0.68187966  1.00000000  0.08394396
## Daily.Steps              0.81117060  0.21514012  0.08394396  1.00000000
## Steps                    0.81117060  0.21514012  0.08394396  1.00000000
##                         Steps
## Quality.of.Sleep  -0.04366560
## GenderBI          -0.02720559
## Age                0.04924485
## Sleep             -0.07307288
## Physical.Activity  0.81117060
## Stress             0.21514012
## Heart.Rate         0.08394396
## Daily.Steps        1.00000000
## Steps              1.00000000
```

Pearson r analysis writeup

For this assignment, a sleep-health dataset was analyzed by G5 in order to identify correlational effects of independent variables (age, sleep duration (hours/night), physical activity level (1-100), stress level (1-10), heart rate, daily steps (thousands), gender) on a selected dependent variable (quality of sleep).

R analysis included only subjects belonging to Generation Y (Millennial) and Generation X. These are individuals aged >=30 years as of 2026. Data cleanup performed by checking the dataset for duplicated and missing values.

Using a correlation matrix, the Pearson correlation coefficients between the IVs and DV were identified.

Analysis found that quality of sleep is negatively correlated with gender (r=-0.27), stress (r=-0.90), heart rate (r=-0.62), and steps/daily steps (r=-0.04), and positively correlated with age (r=0.42), sleep duration (r=0.89), and physical activity (r=0.13).

(Regarding gender, value increases were correlated with decreased quality of sleep. This indicates that being female (gender value=0) correlates to lower quality of sleep.)

These results indicate that as stress level, heart rate, and steps per day increase, quality of sleep decreases. Considering the scale of correlation coefficient, steps has very low negative correlation (0<r<=-0.19), gender has high negative correlation (-0.6<r<=-0.79), and stress has very high negative correlation (-0.8<r-1.0).

Results also indicate that as age, sleep duration, and physical activity levels increase, quality of sleep also increases. Considering the correlational scale, age has a moderate correlation (0.4<r<=0.59), sleep duration has a very high correlation (0.8<r1.0), and physical activity has a very low correlation (0<r<=0.19).

Overall, these findings indicate that stress and sleep duration are most strongly correlated with quality of sleep, with stress being a negative factor, and sleep duration being a positive factor.

```r
#16.    Find the correlation between any 2 variables by applying least square
# linear regression model.
sleep_cor = cor(sleepDF$Quality.of.Sleep, sleepDF$Stress)
sleep_cor
```

```
## [1] -0.9013346
```

Hence we are performing the correlation cal on STRESS

#Ready for Training

```r
#17.    Create a training set using a random number generator engine.
set.seed(4033)
sleepDF_train = sleepDF %>% sample_frac(0.8, replace = FALSE)
head(sleepDF_train)
```

```
##   Quality.of.Sleep GenderBI Age Sleep Physical.Activity Stress Heart.Rate
## 1                7        1  30   7.9                75      6         70
## 2                6        1  43   6.5                45      7         72
## 3                7        0  39   6.9                50      6         75
## 4                8        0  37   7.2                60      4         68
## 5                5        0  30   6.4                35      7         78
## 6                6        1  32   6.2                30      8         72
##   Daily.Steps Steps
## 1        8000   8.0
## 2        6000   6.0
## 3        5500   5.5
## 4        7000   7.0
## 5        4100   4.1
## 6        5000   5.0
```

To avoid data leakage splitting for training and predictive analytics usually is done before analysis. However, this was is out of scope of current analysis. Splitting was performed to fulfil assignment requirements.